1MT495 Current Issues in Applied International Economics: Fundamental Theories Explaining Exchange Rates in International Finance

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Overview

Standardized Notation

To ensure consistency, we use the following notation throughout:

- S_t: Nominal exchange rate (domestic currency per foreign unit). $\uparrow S_t$ $=$ Depreciation of domestic currency.
- $s_t = \ln S_t$: Log nominal exchange rate.
- $\Delta s_t \approx$ Percentage change (depreciation rate).
- $Q_t = S_t \cdot \frac{P_t^*}{P_t}$: Real exchange rate.
- $q_t = \ln Q_t = s_t + p_t^* - p_t$: Log real exchange rate.
- i_t, i_t^*: Domestic and foreign nominal interest rates.
- r_t, r_t^*: Domestic and foreign real interest rates.
- P_t, P_t^*: Domestic and foreign price levels ($p_t = \ln P_t$).
- π_t, π_t^*: Domestic and foreign inflation rates.
- m_t, m_t^*: Log money supplies.
- y_t, y_t^*: Log outputs/incomes.
- $E_t[\cdot]$: Expectation conditional on time t information.

Definition and Types of Exchange Rates

Definitions

- **Nominal Exchange Rate (S_t)**: Price of one currency in terms of another (e.g., USD/EUR).
- **Real Exchange Rate (Q_t)**: Adjusts for price differences.

$$Q_t = S_t \cdot \frac{P_t^*}{P_t}$$

- **Nominal Effective Exchange Rate (NEER)**: Trade-weighted average against a basket of currencies.

$$NEER_t = \prod_{j=1}^{n} (S_{j,t})^{w_j} \quad \text{where } \sum w_j = 1$$

- **Real Effective Exchange Rate (REER)**: NEER adjusted for relative inflation.

Overview of Key Theories

We will cover theories based on fundamentals:

- **Purchasing Power Parity (PPP)**: Long-run price level linkage.
- **Uncovered Interest Parity (UIP)**: Interest rates and expectations.
- **International Fisher Effect (IFE)**: Inflation and interest differentials.
- **Monetary Model**: Money supply and demand fundamentals.
- **BOP & Mundell-Fleming**: IS-LM in open economies.
- **Dornbusch Overshooting**: Sticky prices and dynamics.

Determinants of Exchange Rates

Exchange rates determined in FOREX markets by arbitrage, speculation, and fundamentals.

- **Supply and Demand**: Trade (goods/services), Capital flows (FDI, portfolio), Central bank interventions, Market speculation.
- **Macro Fundamentals**: Inflation, interest rates, eonomic growth, productivity.
- **Microstructure**: Order flow, liquidity, high-frequency trading.
- **Behavioral Factors**: Sentiment, herding, overreaction, political stability (Volatility: fat tails, clustering, leverage effect).

Key Theories Overview

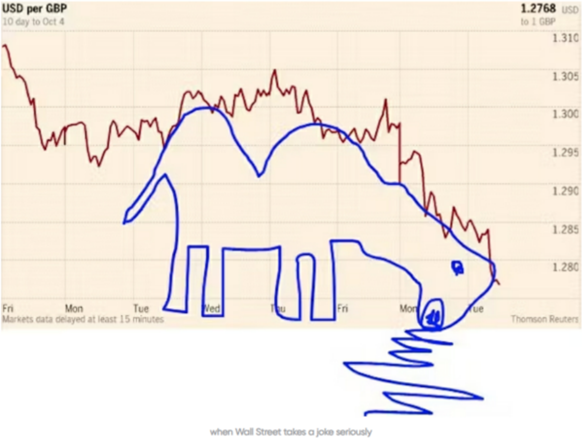
We will cover (theories based on fundamentals):

- Purchasing Power Parity (PPP): Long-run price level linkage.
- Uncovered Interest Parity (UIP): Interest rates and expectations.
- International Fisher Effect (IFE): Inflation and interest differentials.
- Monetary Model: Money supply and demand fundamentals.
- Other models: Balance of Payments (BOP), Mundell-Fleming.
- Forward-looking properties: Rational expectations and dynamics.
- Implications for forecasting: Empirical challenges and methods.

These theories provide frameworks for understanding long-run and short-run exchange rate behavior.

What We Will NOT Cover

We will not cover chartist/technical analysis.



"The Vomiting Camel Pattern"

Uncovered Interest Parity (UIP): Concept

The Core Intuition

The domestic interest rate must equal the foreign interest rate plus the expected depreciation of the domestic currency.

Assumptions:

- Perfect capital mobility
- No transaction costs
- Risk-neutral investors (No risk premium, No default risk)
- Rational expectations

UIP: Derivation (1/2)

Consider an investor with 1 unit of **Domestic Currency**. They have two options:

Option 1: Invest Domestically

$$\text{Return} = 1 + i_t$$

Option 2: Invest Abroad

1. Convert to foreign currency at spot rate S_t (Domestic/Foreign). Amount: $1/S_t$.
2. Invest at foreign rate i_t^*. Amount: $\frac{1}{S_t}(1 + i_t^*)$.
3. Convert back at expected future spot rate $E_t[S_{t+1}]$.

$$\text{Expected Return} = \frac{E_t[S_{t+1}]}{S_t}(1 + i_t^*)$$

UIP: Derivation (2/2)

No Arbitrage Condition:

$$1 + i_t = (1 + i_t^*) \frac{E_t[S_{t+1}]}{S_t}$$

Taking natural logs (approx. $\ln(1 + x) \approx x$):

$$\ln(1 + i_t) = \ln(1 + i_t^*) + \ln(E_t[S_{t+1}]) - \ln(S_t)$$

Standard UIP Equation

$$i_t - i_t^* = E_t[\Delta s_{t+1}]$$

Where $E_t[\Delta s_{t+1}]$ is the expected rate of depreciation.

UIP: Timing and Implications

- **Timing:** UIP links current interest differentials to *future* exchange rate expectations.
- **Mechanism:** If Domestic i_t rises relative to Foreign i_t^*:

$$E_t[\Delta s_{t+1}] = i_t - i_t^* > 0$$

  Market expects the currency to **depreciate** in the future.
- **Overshooting (Current Effect):** To generate expected future depreciation, the currency **appreciate immediately** (Spot S_t falls).
- **Example (Fed Hike):** USD appreciates instantly (jump), then is expected to weaken over time to equalize returns.
- Short-run appreciation dominates (Higher i_t attracts immediate capital inflows, causing current appreciation (spot s_t falls); long-run puzzle violates pure expectation.

UIP vs. Covered Interest Parity (CIP)

Covered Interest Parity (CIP): Uses Forward contracts.

$$f_{t,T} - s_t = (i_t - i_t^*)(T - t)$$

- Risk-free arbitrage (hedged).
- Holds tightly empirically (except post-2008 due to basis spreads).

Uncovered Interest Parity (UIP): Uses Expectations.

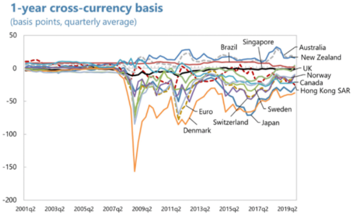
$$E_t[s_{t+T}] - s_t = (i_t - i_t^*)(T - t)$$

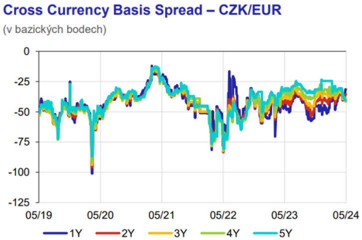
- Risky (unhedged).
- Difference implies Risk Premium (rp_t):

$$f_{t,T} - E_t[s_{t+T}] = rp_t$$

Deviations from CIP

Post-2008, CIP deviations appear as the **Cross Currency Basis**.





UIP: Empirical Evidence

The Fama Regression (Forward Premium Puzzle): Test equation:

$$\Delta s_{t+1} = \alpha + \beta(i_t - i_t^*) + \epsilon_{t+1}$$

- **UIP Prediction:** $\alpha = 0, \beta = 1$.
- **Empirical Finding:** $\beta < 1$ (often even negative).
- **Implication:** High interest rate currencies tend to *appreciate* rather than depreciate in the short run (Forward bias).

Explanations:

- **Carry Trade:** Profiting from the failure of UIP (Borrow Low i^*, Lend High i).
- **Risk Premia:** Time-varying risk (rp_t) absorbs the differential.
- **Rare disasters**: Peso problem small probability large depreciation.
- **Limits to Arbitrage:** Funding constraints and noise traders, Over-extrapolation of trends (Behavioral factors).
- **Long-Run:** β approaches 1 over horizons of 5-10 years (Chinn-Meredith, 2004).

UIP: Implications

- High interest rate currencies expected to depreciate.
- Monetary transmission: Rate hikes appreciate currency.
- Links financial markets to exchange rates.
- Incorporates market expectations via interest differentials.
- Carry strategies (basis for FX trading strategies): Borrow low-i, lend high-i.
- Crisis prediction: Deviations signal imbalances (e.g., if i_t rises, expected depreciation).

UIP: Criticisms

- Empirical failure (UIP puzzle).
- Ignores time-varying risk premiums.
- Market inefficiencies, capital controls.
- Poor short-run (out-of-sample) forecasting vs. random walk.

UIP: Time-Varying Risk Premium

Exchange rates often behave like random walks. We can augment UIP to include a risk premium (rp_t):

$$i_t - i_t^* = E_t[\Delta s_{t+1}] + rp_t$$

- rp_t varies with volatility, liquidity, or consumption risk.
- If rp_t is volatile and correlated with interest differentials, it explains the regression bias ($\beta < 0$).
- **Asset Pricing View:** S_t is the discounted value of future fundamentals; a time-varying discount rate causes S_t volatility.

Purchasing Power Parity (PPP): Concept

PPP asserts that identical goods should cost the same across countries when priced in a common currency (Law of One Price).

Notation:

$$P_t = S_t \times P_t^*$$

(Domestic Price = Exchange Rate × Foreign Price)

Versions:

- **Absolute PPP:** Levels equate ($S_t = P_t / P_t^*$).
- **Relative PPP:** Changes equate ($\Delta s_t = \pi_t - \pi_t^*$).

Assumptions: No transport costs, no trade barriers, perfect competition, homogeneous goods.

PPP: Derivation of Absolute PPP

Consider a basket of goods with price level P_d domestically, P_f abroad. By LOOP aggregated:

$$P_t = S \cdot P_t^* \implies S_t = \frac{P_t}{P_t^*}$$

In log form:

$$s_t = p_t - p_t^*$$

Deviations $q_t = s_t - (p_t - p_t^*)$ is the real exchange rate; PPP implies $q_t = 0$.
Empirically, q_t mean-reverts slowly.
Alternatively: From LOOP for a basket of goods:

$$P_t = \sum w_i P_{i,t} = S_t \sum w_i P_{i,t}^* = S_t P_t^*$$

assuming identical weights w_i.

PPP: Derivation of Relative PPP

From absolute PPP at t and t+1:

$$S_t = \frac{P_t}{P_t^*}, \quad S_{t+1} = \frac{P_{t+1}}{P_{t+1}^*}$$

Change:

$$\frac{S_{t+1}}{S_t} = \frac{P_{t+1}/P_t}{P_{t+1}^*/P_t^*} = \frac{1+\pi_{t+1}}{1+\pi_{t+1}^*}$$

Approximating:

$$\frac{S_{t+1}-S_t}{S_t} \approx \pi_{t+1} - \pi_{t+1}^*$$

Or in logs:

$$\Delta s_{t+1} = \pi_{t+1} - \pi_{t+1}^*$$

This links inflation differentials to exchange rate changes.

Sources of Real Exchange Rate Volatility: Derivation 1/2

The Real Exchange Rate (q_t) and Price Levels:

$$q_t = s_t + p_t^* - p_t$$

Using the price index $p_t = \alpha p_{T,t} + (1-\alpha)p_{N,t}$ (and similarly for foreign with weight α^*):

Step 1: Substitute Price Levels

$$q_t = s_t + [\alpha^* p_{T,t}^* + (1-\alpha^*)p_{N,t}^*] - [\alpha p_{T,t} + (1-\alpha)p_{N,t}]$$

Step 2: Rearrange to Isolate Components By adding and subtracting terms, we can decompose q_t into three distinct parts:

$$q_t = \underbrace{(s_t + p_{T,t}^* - p_{T,t})}_{\text{1. LOOP Deviations}} + \underbrace{[\alpha(p_{T,t} - p_{N,t}) - \alpha^*(p_{T,t}^* - p_{N,t}^*)]}_{\text{2.+3. Non-Tradable Price Movements, Changes in Weights}}$$

The Three Sources of Volatility:

1. **Tradable Price Movements:** Deviations from LOOP (e.g., transport costs, pricing-to-market).
2. **Non-Tradable Price Movements:** Changes in internal relative prices ($p_T - p_N$), often driven by productivity.
3. **Changes in Weights (α, α^*):** Structural changes in the consumption basket (if α shifts, q_t changes even if prices remain constant).

PPP: The Balassa-Samuelson Effect (Concept)

Proposed by Bela Balassa (1964) and Paul Samuelson (1964).
Observation: Richer countries have higher aggregate price levels (Real Appreciation).
Mechanism:

1. Productivity growth is higher in the **Tradable** sector (a_T) for rich countries.
2. High a_T leads to higher wages (W) in the whole economy (labor mobility).
3. High W raises costs/prices in the **Non-Tradable** sector (where productivity a_N is low).
4. Result: Higher overall Price Level (P_t).

Assumption: $a_T > a_T^*$ (Domestic is Rich), but $a_N \approx a_N^*$.

Assumptions: Two sectors (tradables T, non-tradables N); labor mobility; LOP for tradables.

1. Sectoral Prices (from Production): Prices are driven by unit labor costs (w = wage, a = productivity, Prices in levels: $P_T = \frac{W}{A_T}$, $P_N = \frac{W}{A_N}$).

$$p_T = w - a_T \quad ; \quad p_N = w - a_N$$

2. Internal Relative Prices:

$$p_N - p_T = (w - a_N) - (w - a_T) = a_T - a_N$$

(Higher tradable productivity makes non-tradables relatively expensive).

3. International Link (LOOP): Assume LOOP holds for Tradables:

$$s_t = p_T - p_T^*$$

Balassa-Samuelson: Derivation (2/2)

4. Real Exchange Rate Impact: Recall the definition of q_t (using aggregate prices):

$$q_t = s_t + p_t^* - p_t$$

Substitute $p_t = p_T + (1-\alpha)(p_N - p_T)$ and $s_t = p_T - p_T^*$:

$$q_t = (p_T - p_T^*) + [p_T^* + (1-\alpha)(p_N^* - p_T^*)] - [p_T + (1-\alpha)(p_N - p_T)]$$

Simplify:

$$q_t = (1-\alpha)[(p_N^* - p_T^*) - (p_N - p_T)]$$

Substitute productivity differentials ($p_N - p_T = a_T - a_N$):

$$q_t = (1-\alpha)[(a_T^* - a_N^*) - (a_T - a_N)]$$

Balassa-Samuelson: Implication

Final Equation:

$$q_t = (1-\alpha)[(a_T^* - a_N^*) - (a_T - a_N)]$$

If Domestic is the "Rich" country ($a_T > a_T^*$ and $a_N \approx a_N^*$):

- The term $-(a_T - a_N)$ is large and negative.
- Therefore, $q_t < 0$.

Conclusion: A country with higher productivity in tradables will experience a **Real Appreciation** (Lower q_t, Higher P_t). This explains why the Real Exchange Rate is not stationary across countries with different development levels.

PPP: Empirical Evidence & Implications

Evidence:

- **Short Run:** PPP fails. Volatile s_t disconnects from sticky P_t.
- **Long Run:** PPP holds better. Deviations (q_t) mean-revert with a half-life of 3–5 years. Outperforms random walk in very long horizons.
- **Big Mac Index:** A popular measure of PPP over/undervaluation.

Implications:

- Functions as a long-run anchor for exchange rates.
- **Monetary Approach:** High inflation differential ($\pi > \pi^*$) leads to depreciation ($\Delta s > 0$). (Better for hyperinflation scenarios.)
- Relative PPP can be extended to expectations:
  $E(\Delta s) = E(\pi_d) - E(\pi_f)$.

PPP: Criticisms

- Short-run failures due to sticky prices.
- Basic version do not account for non-traded goods (Balassa-Samuelson: Productivity differences inflate non-tradables, segmentation).
- Basic version also lacks transaction costs, tariffs.
- Weak empirical support in short run; better long run.

PPP: Empirical Evidence

- Long-run forecasts: Use inflation differentials to predict equilibrium rates.
- Unit root tests: ADF, PP on $q_t = s_t + p_t^* - p_t$; reject non-stationarity for mean-reversion.
- Testing: Cointegration between s_t, p_t, p_t^*.
- Half-life: $h = \frac{\ln(0.5)}{\ln(\rho)}$, where $q_t = \rho q_{t-1} + \epsilon_t$; typically 3-5 years.
- Guides long-run equilibrium; useful for policy (e.g., inflation targeting).

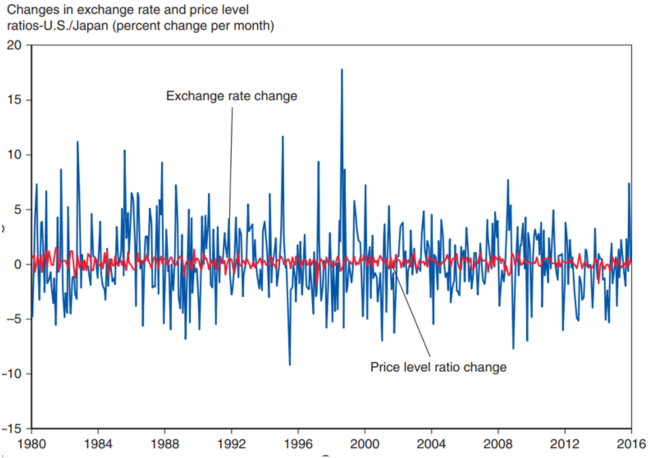
PPP: Case Study IKEA

Table 1. *European Prices (1998, in US$)*

	Alg Mirror [1]	Krabb Mirror [2]	Ratio of Alg to Krabb [3]	Guldros Mirror [4]
Netherlands	20	20	1.03	100
United Kingdom	25	30	0.83	114
Belgium	22	27	0.80	110
Switzerland	19	27	0.72	68
Finland	13	19	0.70	93
Spain	22	33	0.65	110
France	21	33	0.64	99
Sweden	15	23	0.64	91
Italy	22	43	0.52	78
Austria	24	48	0.49	114
Norway	12	26	0.46	79
Germany	22	51	0.44	96
Denmark	12	30	0.40	104

Notes: Alg: square mirror, 45 × 60 cm, 2 tiles. *Guldros*: round mirror, 59 × 78 cm, beveled glass. *Krabb*: designer mirror, 44 × 40 cm. *Source:* IKEA. Exchange rate: annual average (series rf) from IMF *International Financial Statistics*.

PPP: Volatility



Changes in price levels are less volatile, suggesting that price levels change slowly.

International Fisher Effect (IFE): Concept

IFE states that nominal interest rate differentials reflect expected inflation differentials, implying that expected real returns are equalized across countries.

Underlying Components:

- **Domestic Fisher Equation:** Links nominal rate (i) to expected real rate (r):

$$i_t = r_t + E_t[\pi_{t+1}]$$

- **IFE Core Hypothesis:** Assumes real rates are equalized ($r_t = r_t^*$) due to global capital market integration and rational expectations.

$$i_t - i_t^* = E_t[\pi_{t+1} - \pi_{t+1}^*]$$

Assumptions:

- Relative Purchasing Power Parity (PPP) holds.
- Real interest rate parity holds tightly ($r_t = r_t^*$).
- No country-specific risk premia.

IFE: Derivation and Link to PPP

The IFE is derived by taking the difference between the domestic and foreign Fisher equations:

1. Fisher Differentials:

$$i_t - i_t^* = (r_t - r_t^*) + E_t[\pi_{t+1} - \pi_{t+1}^*]$$

2. IFE Simplification: Assuming **Real Interest Rate Parity ($r_t = r_t^*$)**, the real rate differential drops out:

$$i_t - i_t^* = E_t[\pi_{t+1} - \pi_{t+1}^*]$$

3. Link to Expected Depreciation (PPP): From the expected form of **Relative PPP**, expected inflation differentials equal expected exchange rate depreciation:

$$E_t[\pi_{t+1} - \pi_{t+1}^*] = E_t[\Delta s_{t+1}]$$

Final IFE/UIP Equivalence: IFE implies the interest differential predicts currency movement:

$$i_t - i_t^* = E_t[\Delta s_{t+1}]$$

This equation is mathematically identical to the Uncovered Interest Parity

IFE: Derivation and Linkages

The general relationship without assuming real rate parity ($r_t = r_t^*$):

$$i_t - i_t^* = (r_t - r_t^*) + E_t[\pi_{t+1} - \pi_{t+1}^*]$$

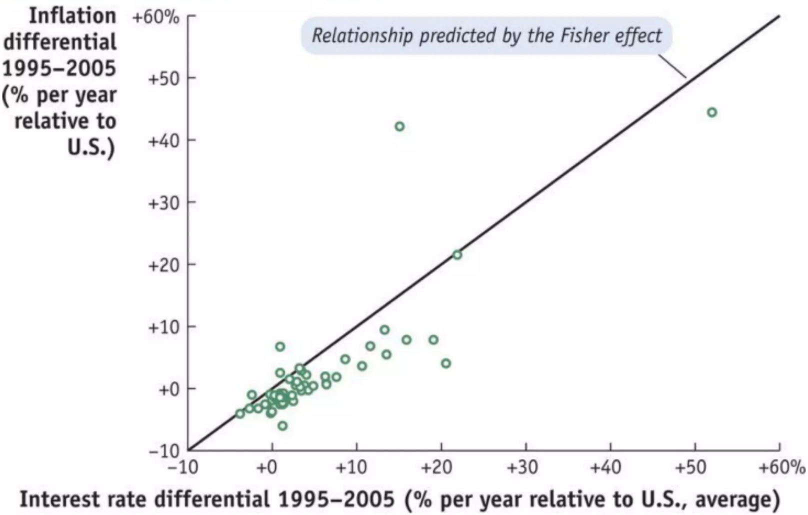
- **Real Interest Rate Parity (RIP):** $r_t - r_t^* = E_t[\Delta q_{t+1}]$. RIP relates real rate differentials to expected changes in the real exchange rate (q_t).
- **IFE Implies UIP:** If both IFE ($r_t = r_t^*$) and expected Relative PPP ($E_t[\Delta s] = E_t[\pi - \pi^*]$) hold, then UIP must also hold.

Ex Post Realized Differential: The difference between the realized nominal differential and the realized inflation differential:

$$(i_t - i_t^*) - (\pi_{t+1} - \pi_{t+1}^*) = (r_t - r_{t+1}^*) + (\pi_{t+1}^e - \pi_{t+1}) - (\pi_{t+1}^{*e} - \pi_{t+1}^*)$$

This deviation equals the real rate differential plus realization errors in inflation forecasts.

IFE: Empirical Evidence



IFE: Implications and Criticisms

Implications:

- **Forecasting:** Currencies in countries with higher nominal interest rates are expected to depreciate due to higher underlying inflation expectations.
- **Monetary Policy:** Tight monetary policy that tames inflation expectations (lower $E_t[\pi]$) should lead to currency appreciation.
- **Testing:** IFE is tested empirically by regressing exchange rate changes on past inflation differentials or nominal interest differentials.

Criticisms:

- **Real Rate Differentials:** Real rates (r, r^*) differ widely due to country-specific factors (fiscal policy, risk, taxes, liquidity, market segmentation), violating the core IFE assumption.
- **PPP Failure:** IFE relies on PPP holding, but PPP fails dramatically in the short run (sticky prices, non-tradables).
- **Empirical Evidence:** Like UIP, IFE generally fails in empirical tests, particularly over short horizons.

The Monetary Model: Concept

Core Idea

Exchange rates are viewed as **asset prices** determined by the relative supply and demand for money in two countries.

Key Assumptions:

- Purchasing Power Parity (PPP) holds continuously.
- Stable Money Demand functions.
- Uncovered Interest Parity (UIP) holds.

Two Main Versions:

1. **Flexible-Price Monetary Model (FPMM):** Prices adjust instantly. Exchange rate determined by money market equilibrium and PPP.
2. **Sticky-Price (Dornbusch):** Prices are sticky in the short run, allowing for exchange rate *overshooting*.

FPMM: Derivation (1/2)

1. Money Market Equilibrium (Log-linear form):

$$\begin{aligned} \text{Domestic:} \quad & m_t - p_t = \phi y_t - \lambda i_t \\ \text{Foreign:} \quad & m_t^* - p_t^* = \phi y_t^* - \lambda i_t^* \end{aligned}$$

Where ϕ is income elasticity and λ is semi-elasticity of interest.

2. Purchasing Power Parity (PPP):

$$s_t = p_t - p_t^*$$

3. Subtract Foreign from Domestic Money Demand:

$$(m_t - m_t^*) - (p_t - p_t^*) = \phi(y_t - y_t^*) - \lambda(i_t - i_t^*)$$

FPMM: Derivation (2/2)

4. Solve for the Exchange Rate (s_t): Substituting $s_t = p_t - p_t^*$ into the previous equation:

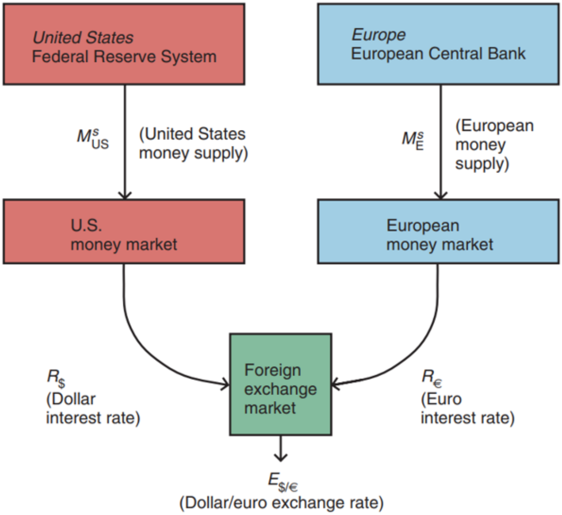
The Fundamental Monetary Equation

$$s_t = (m_t - m_t^*) - \phi(y_t - y_t^*) + \lambda(i_t - i_t^*)$$

Implications (Comparative Statics):

- **Money Supply ($\uparrow m_t$):** Depreciation ($\uparrow s_t$). More money chases same goods $\rightarrow$ Inflation $\rightarrow$ Depreciation (PPP).
- **Real Income ($\uparrow y_t$): Appreciation** ($\downarrow s_t$). Higher income $\rightarrow$ Higher money demand $\rightarrow$ Currency strengthens.
- **Interest Rate ($\uparrow i_t$): Depreciation** ($\uparrow s_t$). Higher opportunity cost $\rightarrow$ Lower money demand $\rightarrow$ Excess supply of money causes depreciation.

Monetary Model Visualization



The Forward-Looking Monetary Model

What if we incorporate Rational Expectations via UIP?

Recall the fundamental equation: $s_t = \text{Fundamentals}_t + \lambda(i_t - i_t^*)$.

Substitute UIP ($i_t - i_t^* = E_t[s_{t+1}] - s_t$):

$$s_t = (m_t - m_t^* - \phi(y_t - y_t^*)) + \lambda(E_t[s_{t+1}] - s_t)$$

Rearranging for s_t yields the **Present Value** form:

$$s_t = \frac{1}{1+\lambda} f_t + \frac{\lambda}{1+\lambda} E_t[s_{t+1}]$$

Where $f_t = (m_t - m_t^*) - \phi(y_t - y_t^*)$ represents current economic fundamentals.

Solution via Iteration

Solving forward recursively shows s_t is the PV of *all* future fundamentals:

$$s_t = \sum_{k=0}^{\infty} \left(\frac{\lambda}{1+\lambda}\right)^k E_t[f_{t+k}]$$

Empirical Evidence and Extensions

Successes:

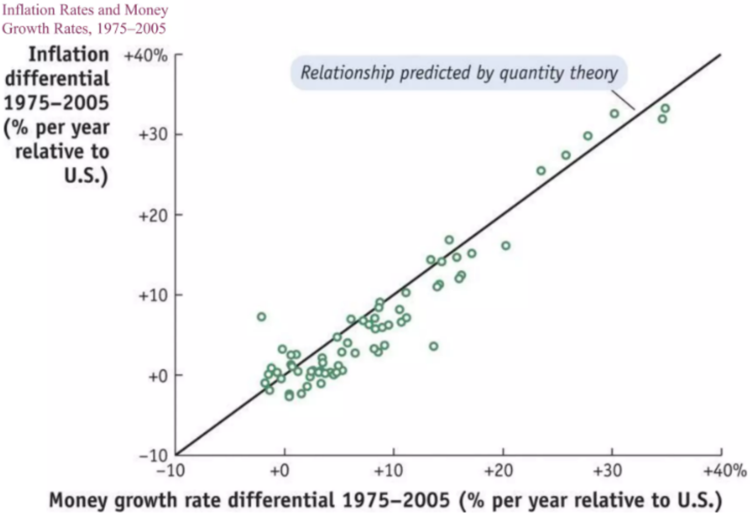
- Performs well in **Hyperinflation** scenarios (e.g., 1920s Germany).
- Captures long-run trends where monetary aggregates dominate.

Failures & Challenges:

- **Meese-Rogoff Puzzle:** Poor out-of-sample forecasting in the short run (Random Walk beats the model).
- **Instability:** Unstable money velocity and demand functions.

Key Insight: Because s_t depends on $E_t[\text{future } f]$, "News" about future policy affects the spot rate **instantly**.
Extensions: Taylor rule for i instead of money supply.

Empirical Evidence (Hyperinflation)



The Trade Balance (Elasticities) Model

Overview

A partial equilibrium approach analyzing how changes in the exchange rate affect the Trade Balance ($TB = X - M$) through relative price changes.

Scope & Focus:

- **Focus:** Analyzes adjustments in the **Current Account** given an exogenous change in the exchange rate.
- **Regime Relevance:**
  - **Fixed Rates:** Analyzing the efficacy of a discrete **Devaluation**.
  - **Floating Rates:** Analyzing how market depreciation might correct a deficit.

Key Assumptions:

- Partial equilibrium (Income Y and Prices P are held constant).
- Supply is infinitely elastic (demand-driven model).
- Small open economy.

The Mechanics of Depreciation

The Equation: Expressing the Trade Balance in domestic currency terms:

$$TB = \underbrace{P_X \cdot Q_X}_{\text{Export Value}} - \underbrace{S_t \cdot P_M^* \cdot Q_M}_{\text{Import Value}}$$

The Effect of Depreciation ($S_t \uparrow$):

1. **Valuation Effect (Price):** Immediate increase in the domestic price of imports ($S_t P_M^*$).
2. **Volume Effect (Quantity):**
   - Exports become cheaper abroad $\rightarrow Q_X \uparrow$.
   - Imports become expensive at home $\rightarrow Q_M \downarrow$.

The net effect depends on which dominates: the higher cost of imports or the improved volume of trade.

Under what conditions does a depreciation actually improve the Trade Balance?

The Marshall-Lerner Condition

Assuming an initial balanced trade ($TB = 0$), a real depreciation improves the trade balance if and only if the sum of export and import demand elasticities is greater than 1:

$$|\epsilon_X| + |\epsilon_M| > 1$$

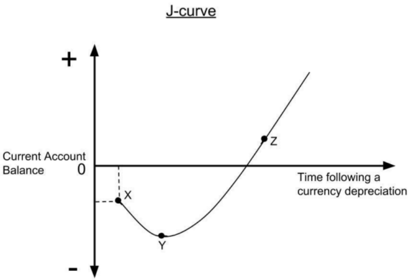
Intuition:

- If demand is **inelastic** (< 1), the volume of trade doesn't change much. The higher cost of imports ($S \uparrow$) dominates, and TB **worsens**.
- If demand is **elastic** (> 1), the volume adjustment ($Q_X \uparrow, Q_M \downarrow$) outweighs the price increase.

Dynamics: The J-Curve Effect

Empirically, trade elasticities are lower in the short run (contracts, habits) than in the long run.

1. **Short Run (The Dip):**
   - $|\epsilon_X| + |\epsilon_M| < 1$.
   - Quantities are sticky.
   - Import costs rise immediately due to $S \uparrow$.
   - **Result:** TB Worsens.
2. **Long Run (The Recovery):**
   - Consumers switch to local goods.
   - ML Condition holds.
   - **Result:** TB Improves.



Implications and Criticisms

Implications

- **Policy Lag:** Devaluation (or depreciation) may cause pain before gain (J-Curve).
- **Feedback Loop:** Large deficits should theoretically cause depreciation, which eventually corrects the deficit (mean reversion).

Major Criticisms (Flow vs. Asset Approach):

- **Partial Equilibrium:** Ignores Income effects. If exports rise, Y rises, which increases import demand (dampening the TB improvement).
- **Pass-Through:** Depreciation does not always translate 1:1 into prices (pricing-to-market).
- **Capital Flows Dominate:** This is a "Flow" model. In reality, exchange rates are often driven by the "Gross Flows" , which can move S_t in the opposite direction of trade flows.

The Mundell-Fleming Model (IS-LM-BP)

Concept

The "Workhorse (Textbook) Model" of open economy macroeconomics. It extends the closed-economy IS-LM framework to include the Balance of Payments (BP).

Crucial Assumptions:

- **Small Open Economy:** Cannot influence world prices (P^*) or interest rates (i^*).
- **Sticky Prices:** P is fixed in the short run.
- **Demand Driven:** Output determined by aggregate demand.

Note on Interest Rates (i vs r)

Because prices are fixed (inflation $\pi = 0$), the **Nominal Interest Rate (i)** equals the **Real Interest Rate (r)**. We will use i to denote the rate on the axes, linking the Money Market to the FX market.

The Impossible Trinity

The Policy Trilemma

A country cannot simultaneously maintain all three:

1. A **Fixed Exchange Rate** (Stability)
2. **Free Capital Mobility** (Integration)
3. **Independent Monetary Policy** (Autonomy)

Mundell-Fleming shows us exactly why one must be sacrificed.

Derivation: The Open Economy IS Curve

Goods Market Equilibrium:

$$Y = C(Y - T) + I(i) + G + NX(Q, Y, Y^*)$$

Components:

- $I(i)$: Investment depends on the interest rate (Cost of borrowing).
- NX: Net Exports depend on the Real Exchange Rate (Q) and Income.
- $Q = \frac{SP^*}{P}$: Real Exchange Rate. (With fixed P, P^*, changes in S drive Q).
- *Notation:* $\uparrow S =$ Depreciation (Domestic currency buys less foreign).

Linear IS Equation:

$$Y = \frac{A_0}{k} - \frac{b}{k} i + \frac{\delta}{k} S$$

(Where A_0 is autonomous spending and k is the multiplier).

Slope and Shifts:

- **Slope:** Downward sloping in (Y, r) space (higher $r \rightarrow$ lower I).
- **Shifts Right if:** $\uparrow G$ (Fiscal expansion) or $\uparrow s$ (Depreciation boosts exports).

Derivation: The Money Market (LM Curve)

Money Market Equilibrium:

$$\frac{M}{P} = L(Y, i)$$

Mechanics:

- **Supply:** Fixed by Central Bank ($\bar{M}$) and sticky prices ($\bar{P}$). Higher Y raises money demand; i must rise to clear the market.
- **Demand:** Increases with Income (Y) (transaction motive) and decreases with Interest Rate (i) (opportunity cost of holding cash).

Slope and Shifts:

- **Slope:** Upward sloping in (Y, i) space.
- **Shifts Right if:** $\uparrow M$ (Monetary expansion).

Derivation: The Balance of Payments (BP Curve)

The BP curve shows (Y, i) combinations where Balance of Payments $= 0$.

$$BP = \underbrace{CA(S, Y)}_{\text{Current Account}} + \underbrace{FA(i - i^* - E[\Delta s])}_{\text{Financial Account}} = 0$$

The Financial Account & UIP: Capital flows depend on the **Uncovered Interest Parity (UIP)** condition. Investors compare domestic returns (i) against foreign returns (i^*) adjusted for expected depreciation ($E[\Delta s]$).

The Role of Capital Mobility:

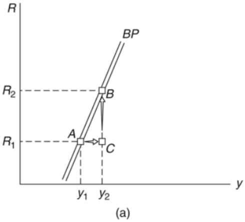
- **Imperfect Mobility:** FA responds slowly to rate differentials. BP is upward sloping (Higher Y worsens CA, requires higher i to attract foreign investments $\uparrow FA$).
- **Perfect Mobility (Standard M-F):** Assets are perfect substitutes. Even a tiny deviation of i from i^* triggers massive flows.

Perfect Mobility Condition

If we assume **Static Expectations** ($E[\Delta s] = 0$), the BP curve is **Horizontal** at $i = i^*$.

Visualizing Capital Mobility

The slope of the BP curve reflects the degree of capital mobility.



Analysis:

- **Steep BP:** Low mobility. Large rate hikes needed to attract capital to offset trade deficits from higher Y.
- **Flat BP:** Perfect mobility. Tiny rate differentials trigger infinite flows.

Equilibrium under Fixed Exchange Rates

The Mechanism: Endogenous Money Supply

- The Central Bank commits to a specific exchange rate $\bar{S}$.
- To maintain $\bar{S}$, the CB must intervene in FX markets.
- **Intervention:** Buying/Selling FX reserves changes the domestic Money Supply (M).

Adjustment:

- If pressure to appreciate (Capital Inflows): CB buys FX, sells Domestic currency $\rightarrow M \uparrow$.
- If pressure to depreciate (Capital Outflows): CB sells FX, buys Domestic currency $\rightarrow M \downarrow$.

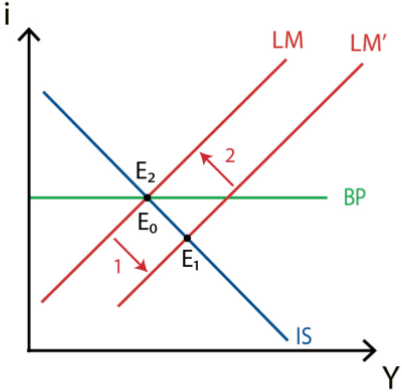
Conclusion: The LM curve becomes endogenous.

Fixed Rates: Monetary Policy (Ineffective)

Scenario: Expansion ($\uparrow M$)

1. CB increases M, shifting **LM right**.
2. Result: $i < i^*$.
3. **Capital Outflows:** Investors sell domestic currency.
4. **Intervention:** To prevent depreciation, CB must buy domestic currency (sell reserves).
5. **Result:** M shrinks, **LM shifts back left**.

Outcome: No change in Y.

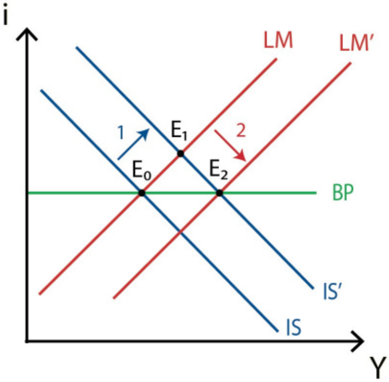


Fixed Rates: Fiscal Policy (Highly Effective)

Scenario: Expansion ($\uparrow G$)

1. $\uparrow G$ shifts **IS right**.
2. Result: $i > i^*$ (due to money demand).
3. **Capital Inflows:** Investors buy domestic currency.
4. **Intervention:** To prevent appreciation, CB sells domestic currency (buys reserves).
5. **Accommodation:** M rises, shifting **LM right**.

Outcome: Y increases significantly.



Equilibrium under Floating Exchange Rates

The Mechanism: Endogenous Exchange Rate

- The Central Bank does not intervene (M is exogenous/fixed).
- The Exchange Rate (S) adjusts to clear the Balance of Payments.

Adjustment:

- If $i > i^*$ (Inflows) $\rightarrow$ Demand for currency $\rightarrow$ **Appreciation** ($\downarrow S$).
- If $i < i^*$ (Outflows) $\rightarrow$ Sale of currency $\rightarrow$ **Depreciation** ($\uparrow S$).

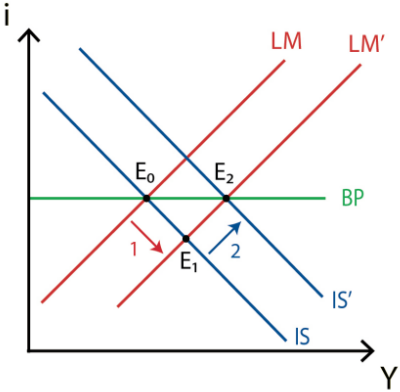
Conclusion: The IS curve shifts due to changes in Net Exports.

Floating Rates: Monetary Policy (Highly Effective)

Scenario: Expansion ($\uparrow M$)

1. CB increases M, shifting **LM right**.
2. Result: $i < i^*$.
3. **Capital Outflows** cause **Depreciation**.
4. Depreciation makes exports cheaper/imports expensive ($\uparrow NX$).
5. $\uparrow NX$ shifts **IS right**.

Outcome: Y increases.

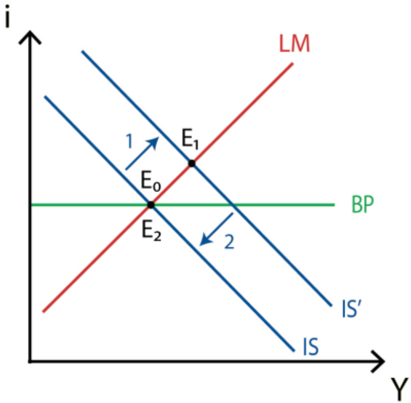


Floating Rates: Fiscal Policy (Ineffective)

Scenario: Expansion ($\uparrow G$)

1. $\uparrow G$ shifts **IS right**.
2. Result: $i > i^*$.
3. **Capital Inflows** cause **Appreciation**.
4. Appreciation hurts exports/helps imports ($\downarrow NX$).
5. $\downarrow NX$ shifts **IS back left**.

Outcome: Full Crowding Out (if perfect mobility).



Implications and Criticisms

Summary of Policy Effectiveness

	Fixed Rate	Floating Rate
Fiscal Policy	Effective	Ineffective
Monetary Policy	Ineffective	Effective

Criticisms Extensions:

- **Static Model:** Ignores dynamics of debt accumulation and expectations (addressed by Dornbusch).
- **Price Rigidity:** Assumes P is fixed; inappropriate for high inflation environments.
- **Empirics:** While the "Trilemma" holds well, the degree of crowding out varies in reality.

The Dornbusch Overshooting Model (1976)

The Big Idea

A dynamic open-economy model that bridges the gap between sticky-price Keynesianism and flexible-price Monetarism.

The Core Puzzle: Why are exchange rates so much more volatile than economic fundamentals (prices, output, money supply)?

Key Assumptions:

- **Small Open Economy** with Perfect Capital Mobility (UIP holds).
- **Rational Expectations** (Agents look forward).
- **Asymmetric Adjustment Speeds:**
  - Asset Markets (s, i): Adjust **instantly**.
  - Goods Markets (p): Adjust **slowly** (Sticky Prices).

Key Equations: The Building Blocks

We use log-linear variables (except interest rates).

1. **Uncovered Interest Parity (UIP):**

$$i_t = i_t^* + E_t[\dot{s}_t]$$

Asset market equilibrium. $\dot{s}$ is expected depreciation.

2. **Money Market (LM):**

$$m_t - p_t = \phi y_t - \lambda i_t$$

Real money supply equals liquidity demand.

3. **Aggregate Demand (IS):**

$$y_t^d = \bar{y} + \delta(q_t - \bar{q}) - \sigma(i_t - \pi_t)$$

Demand depends on Real Exchange Rate (q_t) and Real Interest Rate.

4. **Price Adjustment (Phillips Curve):**

$$\dot{p}_t = \pi(y_t^d - \bar{y})$$

Prices rise when demand exceeds potential output.

Derivation: Long-Run Equilibrium

In the steady state (Long Run), variables are constant ($\dot{s} = 0, \dot{p} = 0$) and output is at potential ($y = \bar{y}$).

1. Interest Rates: From UIP, if $\dot{s} = 0$, then:

$$\bar{i} = i^*$$

2. Prices and Exchange Rates (Neutrality): From Money Demand ($\bar{m} - \bar{p} = \phi\bar{y} - \lambda i^*$):

$$\bar{p} = \bar{m} - (\phi\bar{y} - \lambda i^*)$$

Long-Run Neutrality

A permanent increase in Money Supply (Δm) leads to a proportional increase in Prices and the Exchange Rate (PPP for the long-run equilibrium spot rate):

$$\Delta\bar{p} = \Delta\bar{s} = \Delta m$$

Real variables are unchanged in the long run.

Derivation: Short-Run Dynamics

To find the short-run movement, subtract the Long Run (LR) money equation from the Short Run (SR) equation.
SR: $m - p = \phi \bar{y} - \lambda i$ (Assuming y is close to $\bar{y}$ for simplicity)
LR: $m - \bar{p} = \phi \bar{y} - \lambda i^*$
Subtracting LR from SR:

$$-(p - \bar{p}) = -\lambda(i - i^*)$$

$$i - i^* = \frac{1}{\lambda}(p - \bar{p})$$

Substitute into UIP ($i - i^* = E[\dot{s}]$):

$$E[\dot{s}] = \frac{1}{\lambda}(p - \bar{p})$$

This implies that expected depreciation depends on the price level gap.

Shock Analysis: Monetary Expansion (Graphical)

Money shock: Increase m_t lowers i_t, via UIP requires immediate depreciation beyond long-run to expect appreciation.
Overshooting: $\Delta s_t > \Delta \bar{s}$ for permanent money increase.

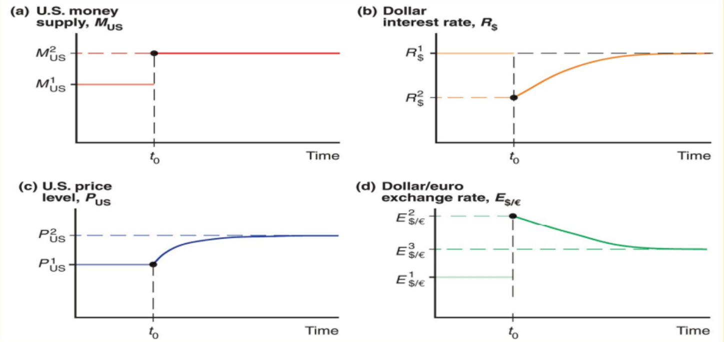


Figure 14-13

Time Paths of U.S. Economic Variables After a Permanent Increase in the U.S. Money Supply

After the money supply increases at t_0 in panel (a), the interest rate (in panel (b)), price level (in panel (c)), and exchange rate (in panel (d)) move as shown toward their long-run levels. As indicated in panel (d) by the initial jump from $E^1_{\$/€}$ to $E^2_{\$/€}$, the exchange rate overshoots in the short run before settling down to its long-run level, $E^3_{\$/€}$.

Mechanism: Why Overshooting Occurs

Consider a permanent monetary expansion ($\uparrow M$).

1. **Liquidity Effect:** Prices (P) are sticky, so Real Money Supply (M/P) rises instantly. Interest rates (i) must **fall** to clear the money market.
2. **UIP Condition:** $i < i^*$. Investors will only hold domestic assets if they expect a capital gain (Appreciation, $\dot{s} < 0$).
3. **The Jump:** We know the currency must depreciate in the Long Run ($\Delta \bar{s} > 0$).
4. **Overshooting:** To satisfy step 2 (expecting appreciation) while satisfying step 3 (long run depreciation), the spot rate s_t must depreciate **instantly and excessively**.

$$s_t > \bar{s}_{new}$$

The currency weakens massively today so it can strengthen back to equilibrium tomorrow.

Relation to Mundell-Fleming

Dornbusch is essentially a "Dynamic Mundell-Fleming" with Rational Expectations.

Similarities:

- Capital mobility: UIP $i = i^* + E[\Delta s]$.
- IS/LM: Shared structures; floating rates: Money expands y, depreciates s.

Feature	**Mundell-Fleming**	**Dornbusch**
Time Horizon	Static (Short Run)	Dynamic (SR transition to LR)
Prices	Fixed (P)	Sticky ($\dot{p}$ adjusts slowly)
Expectations	Static ($E[s] = s_{current}$)	Rational ($E[\dot{s}]$ consistent with model)
Exchange Rate	Determinate level	**Overshooting** (Volatile)

Empirical Criticisms: The "Delayed Overshooting" Puzzle

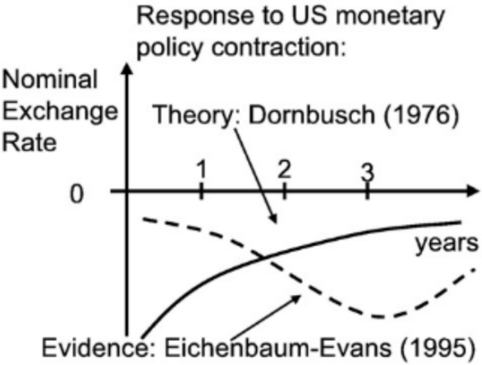
The Theory (Dornbusch):

- A contractionary shock (Rate Hike $i \uparrow$) causes an **instant** peak appreciation.
- UIP holds immediately: The currency is expected to depreciate from the moment of the shock onwards.

The Reality (Eichenbaum & Evans, 1995):

- After a rate hike, the currency appreciates, but the **peak occurs months later**.
- This "hump-shaped" response violates the model's prediction of instant adjustment.

Empirical Criticisms: The "Delayed Overshooting" Puzzle



Source: Alan G. Isaac Lecture notes - Exchange Rate Overshooting

Implication: The model relies on strict UIP, which fails to capture the "Forward Bias" (high interest currencies tend to appreciate further rather than depreciate instantly).

Implications and Criticisms

Implications

- **Volatility:** Explains why exchange rates are more volatile than fundamentals.
- **Liquidity Effect:** Confirms that monetary expansion lowers interest rates only temporarily.
- **Forecasting:** Short-run deviations from PPP are expected and rational.

Criticisms Extensions:

- **Trade Balance:** Originally ignored the Current Account (later added in extensions).
- **Risk Neutrality:** Relies on strict UIP (no risk premium), which fails empirically.
- **Modern Macro:** Precursor to the "New Open Economy Macroeconomics" (NOEM) and the Redux Model (Obstfeld & Rogoff, 1995).

The Forward-Looking Foundation of Exchange Rates

Core Principle: Rational Expectations

Exchange rate models (Monetary, Dornbusch, UIP) are **forward-looking**. Agents use all available information, and the current spot rate (s_t) instantly incorporates expectations ($E_t[\cdot]$) about all relevant future fundamentals.

Key Implications:

- **High Volatility:** Since s_t is the present value of all future fundamentals, a small change in expected future policy or news causes a large, immediate jump in the spot rate.
- **Efficient Markets:** Under Rational Expectations, the spot rate should reflect all available public information, making future changes (Δs_{t+1}) fundamentally **unpredictable** (a random walk) unless new, unexpected **news** arrives.
- **Overshooting (Dornbusch):** The short-run rate must jump past its long-run equilibrium (anchor) to generate the necessary expected return that satisfies UIP.

Rational Expectations and Market Efficiency

Efficiency and Predictability

If the market is **efficient**, the exchange rate already reflects all available information. Therefore, only **unexpected news** can move the rate, making actual changes (Δs) appear random.

Implications for Price Process:

- **Random Walk Tendency:** If the fundamental drivers (f_t) are highly persistent or follow a random walk (I(1) process), the exchange rate itself will closely mimic a random walk.
- **Bubbles and Crashes:** The strong reliance on self-fulfilling expectations means that deviations from fundamentals (rational bubbles) are theoretically possible, contributing to extreme volatility.
- **Unpredictability:** Structural models struggle because they are attempting to forecast the innovation (the news) in the expected stream of fundamentals, which is, by definition, an unpredictable white noise term.

The Challenge: Meese-Rogoff Puzzle

- **Observation:** Since the 1980s, structural, fundamental-based models (Monetary, Dornbusch) have consistently been outperformed by the simple **Random Walk** forecast ($\hat{s}_{t+h} = s_t$) in short to medium horizons.
- **Reason:** The models rely on PPP/UIP holding, but these relationships are weak, unstable, or dominated by risk premia in the short run.

Forecasting Methods (Traditional):

1. **Naive:** Random Walk ($\hat{s}_{t+h} = s_t$).
2. **Fundamental (Long-Run):** Uses **PPP** or Monetary model for trends (e.g., inflation-driven depreciation: $\hat{s}_{t+h} = s_t + h(\bar{\pi} - \bar{\pi}^*)$).
3. **Hybrid (ECM): Error Correction Models (ECM)** combine short-run dynamics with the long-run cointegrated fundamental relationship:

$$\Delta s_t = \alpha(s_{t-1} - \beta' x_{t-1}) + \gamma \Delta x_t + \epsilon_t$$

UIP, Time-Varying Risk Premia (TVRP), and Predictability

The Role of Risk Premia (rp_t): Structural models often fail because they ignore time-varying risk. The true relationship is:

$$i_t - i_t^* = E_t[\Delta s_{t+1}] + \underbrace{rp_t}_{\text{Time-Varying Risk Premium}}$$

Impact of TVRP:

- **UIP Failure:** The TVRP is large and highly variable, dominating the interest differential, which explains why UIP fails empirically (the "forward bias puzzle").
- **Volatility Amplification:** Shocks to risk aversion (changes in rp_t) cause large, immediate jumps in s_t that appear unrelated to macroeconomic fundamentals.

The Engel-West (2005) Present Value Model

This framework unifies the random walk behavior with fundamental dependence.

Model Structure: The spot rate (s_t) is the discounted sum of expected future fundamentals (f_{t+k}):

$$s_t = \sum_{k=0}^{\infty} \left(\frac{1}{1+\lambda} \right)^k E_t[f_{t+k}]$$

(Similar to the FPMM present-value solution)

Key Finding (Engel-West):

- If the discount factor (λ) is near zero, and if fundamentals (f_t) are highly persistent (near random walk or I(1)), then Δs_t will be virtually unpredictable, primarily composed of the unpredictable "news" component.
- **Conclusion:** The low predictive power of fundamentals is not necessarily a flaw of the **theory**, but a consequence of the strong **forward-looking** nature of the market and the high persistence of economic fundamentals.

Advanced Fundamental Models (Taylor Rule & BEER)

Policy-Rule Fundamentals:

- **Taylor Rule Fundamentals:** Using the central bank's interest rate rule (i_t) as the fundamental often improves performance over simple money supply (m_t) models:

$$i_t = r + \pi + \gamma_\pi(\pi - \pi^T) + \gamma_y(y - \bar{y})$$

BEER Model and similar (Behavioral Equilibrium Exchange Rate):

- **Concept:** Moves beyond simple PPP. It estimates the equilibrium real exchange rate ($\bar{q}$) based on medium- to long-term **structural factors** (e.g., net foreign assets, productivity, terms of trade).
- **Advantage:** Provides a more robust long-run anchor for forecasting than simple PPP, acknowledging that the RER equilibrium level can shift.
- **Usage:** $\bar{s} = \bar{q} + \bar{p} - \bar{p}^*$. The disequilibrium $(q - \bar{q})$ is used in an ECM to forecast nominal s_t.

Data Advances and Machine Learning

New Data Sources for Short-Run Forecasts:

- **Order Flow Data:** Captures high-frequency, market-based supply/demand imbalances.
- **Sentiment/News:** Incorporates information from real-time news and social media feeds (e.g., from X).
- **Risk Proxies:** Using options-implied volatility and risk reversals to derive better estimates of TVRP.

Non-Linear and ML Approaches:

- **Machine Learning (ML):** Used to capture non-linear relationships and interactions between fundamentals. Examples include Neural Networks and Random Forests on high-frequency data.
- **Evaluation Shift:** Focus is moving from simple statistical measures (RMSE) to evaluating the **economic value** (e.g., trading profits) and **directional accuracy** of the forecast.

Elective and Ensemble Approaches

Overcoming Instability via Aggregation

- **Parameter Instability:** A major reason for the Meese-Rogoff puzzle is that model parameters (β in ECM) shift over time (regime shifts).
- **Dynamic Model Averaging (DMA):**
  - Approach: Instead of choosing one model, DMA tracks the time-varying probability that each model (e.g., PPP, Monetary, Taylor Rule) is the correct one.
  - Forecast: The final forecast is a weighted average of all competing model forecasts, where weights change dynamically.
  - Benefit: Allows the forecaster to adapt to regime shifts and parameter instability without imposing fixed structure.
- **Forecast Combination (FC):** A simpler, fixed-weight average of various models, based on past performance.

PPP and Long-Horizon Forecasting

PPP as a Long-Run Predictor

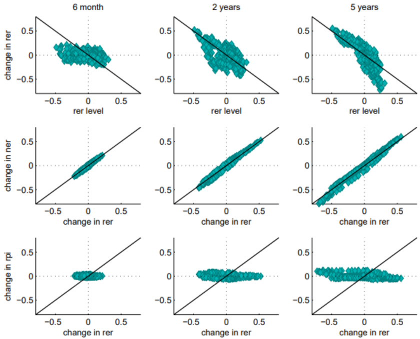
While PPP fails in the short run, the **Real Exchange Rate (RER)** exhibits **mean-reversion** over long horizons (3-5 years). This makes PPP-based models useful for structural long-term planning, particularly when inflation differentials are high.

Forecasting Reality:

- The nominal exchange rate (S) is primarily driven by highly persistent movements in the RER, not by short-term changes in price levels alone.
- Effective long-run forecasting must incorporate the speed at which the RER corrects back toward its long-run mean (the α coefficient in the ECM).

PPP: Forecasting example

Figure 1: Exchange rate regularities for the euro dollar



Notes: Changes for a variable Y over horizon H are expressed as $\Delta y_{t,H} = y_t - y_{t-H}$, where $y_t = \log(Y_t)$. The `rer level` is equal to the deviation of rer_{t-H} from the sample mean $\overline{rer}$.

Nominal exchange rate is driven by movements in real exchange rate not in the price levels. RER is mean reverting at long horizons
Source: Michele Ca' Zorzi, Micha Rubaszek (2018): Exchange rate forecasting on a napkin

Model Frameworks: Long-Run vs. Short-Run

Two Pillars of Exchange Rate Determination

1. **Long-Run (Anchor) Models:** Determined by goods market equilibrium and relative money supplies.
2. **Short-Run (Dynamics) Models:** Determined by financial market equilibrium, expectations, and policy.

The Building Blocks and Their Contribution

Model / Concept	Core Contribution and Interconnection
PPP / IFE	Establishes the **Long-Run Anchor** (mean-reversion). PPP links prices to S. IFE links inflation/interest differentials to expected ΔS.
UIP	The engine of **Short-Run Dynamics**. It determines the spot rate S_t based on the interest differential and the expected future rate $E_t[S_{t+1}]$.
Monetary Models	Integrates money market equilibrium with the long-run PPP anchor. Shows how monetary fundamentals drive S.
Mundell-Fleming	A static, **Sticky-Price** macroeconomic foundation. Illustrates the effectiveness of fiscal/monetary policy under different exchange rate regimes.
Dornbusch Overshooting	The **Synthesis**. Integrates Monetary (LR) and Mundell-Fleming (SR) by allowing sticky prices and perfect capital mobility. Shows how financial market speed leads to **Exchange Rate Volatility**.

Implications for Volatility and Forecasting

Key Puzzles and Modern Explanations

- **High Volatility:** Explained by **Dornbusch** (overshooting) and the **Forward-Looking** nature of the exchange rate (UIP). S_t must instantly jump to reflect all expected future events.
- **Meese-Rogoff Puzzle:** Why do fundamentals fail to beat the Random Walk ($\hat{s}_t = s_{t-1}$)?
- **The Explanation (Engel-West):** Exchange rates are the **Present Value** of future fundamentals. If fundamentals are highly persistent (I(1)), the rate must be a near random walk, making ΔS_t unpredictable "news."
- **TVRP Failure:** The empirical failure of UIP is due to the large, time-varying **Risk Premium** (rp_t) which dominates predictable components.

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